

2.2.2 Readability

پڑھائی پن یہ ہے کہ دوسرا شخص الگورتھم چلائے بغیر اس کا مقصد سمجھ لے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 07 of 24

2.2.2 Readability

LEARN IT

Readability is how easily another person can understand the purpose and logic of an algorithm **without running it**. It is about how clear the text and the flow are.

How to evaluate readability

1. **Naming conventions** — consistent rules for names. Good names make an algorithm easier to search and understand.
Bad: $v = d / t$
Good: `velocity = distance / time`
2. **Self-documentation** — the algorithm explains itself, so you need fewer comments. Meaningful names and clean formatting make the purpose obvious.
3. **Control-flow simplicity** — keep the logic straightforward. Avoid deep nesting (IF inside IF inside IF), limit extra conditions, and keep each function on one task.
4. **Consistency** — one style throughout: indentation, spacing, and casing (`camelCase` or `snake_case`). Then the logic stands out instead of the mess.

Names → Self-doc → Simple flow → Consistent

WORKED EXAMPLE

Worked example: the same formula, two styles.

```
# Hard to read
d = 100
t = 4
s = 20
v = d / t
if v > s:
    if t > 0:
        print(v)
```

OUTPUT
25.0

```
# Readable
distance = 100
time = 4
velocity = distance / time
speed_limit = 20
if time > 0 and velocity > speed_limit:
    print(velocity)
```

OUTPUT
25.0

The second version uses names, avoids an IF nested inside an IF, and keeps one consistent `snake_case` style. You can understand it without running it. Both print `25.0`, but only the second is clear.

REMEMBER IT

Remember readability as **N-S-S-C**: **N**ames, **S**elf-documentation, **S**imple flow, **C**onsistency.

COMMON MISTAKE

Mistake: dumping a comment on every line (`total = total + 1 # add one to total`).

Fix: if the name already says it, skip the comment. Comments should explain **why**, not repeat **what**.

اردو میں سمجھیں

پڑھائی پن یہ ہے کہ دوسرا شخص الگورتھم چلائے بغیر اس کا مقصد سمجھ لے۔ اچھے نام (`velocity = distance / time`) خود بیان (کم تبصرے)، سادہ کنٹرول فلو (گہری IF کے بغیر) اور ایک جیسا انداز (casing اور indentation) اسے آسان بناتے ہیں۔

CHECK YOURSELF

1. Rewrite `v = d / t` with readable names.
2. What is deep nesting, and why is it a readability problem?
3. Name two style habits that support consistency.

Answers: 1. `velocity = distance / time`. 2. IF inside IF inside IF (or similar); it makes the flow hard to follow. 3. Any two of: consistent indentation, spacing, and casing (camelCase or snake_case).

Next lecture: 2.3 Assess Algorithm Efficiency